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Components:

BE HAPPY WITH THE EXPENSIVE CONNECTORS Than The Inexpensive Units

Connectors are one of those components that are not given enough attention just because these are not as complex as microcontrollers or sensors. This interview intends to make engineers understand the significance of selecting the right connector. Ravindra Kumar, country manager at Samtec Inc., speaks with Dilin Anand from EFY

Q. What is your advice to engineers considering a connector for their next project?

A. They should choose one that has a common or standard pitch. Connectors may be structurally simpler than a microcontroller, but selecting the right one is complicated and plays an important role.

Q. Is there a difference between a common and an uncommon connector?

A. The simple difference is that, with a common pitch you can move on tomorrow. Hardware designs are not perpetually defined when the first batch is manufactured. So engineers need to design their product and select the connectors keeping in mind potential future needs and requirements as well as interoperability. It may be possible to select different types of connectors, but we recommend something that is sustainable as part of a design.

Q. Are there any long-term issues that non-standard connectors could cause?

A. Connector vendors have to manage their stock and supplies keeping multiple customers and factory conditions in mind. If you go for a non-standard connector, the vendor might delay supplies later, which can affect shipment of your products. In some cases, this could make or break your product's survival in the market. With standard connectors, you will have backup in the form of other ven-

dors who are able to supply the same product, with footprint compatibility.

Q. How are some components much cheaper than the others?

A. When engineering teams scout for components, they usually end up finding the first lot that is extremely cheap and delivered by a firm importing from countries like China or Taiwan. While the first lot is of high-quality components delivered at what seems like throwaway prices, subsequent lots are of lower quality. The supplier might not figure out the change in quality until defects start rolling in or a product recall is issued.

Q. What kind of quality degradation are we talking about here?

A. Most common is changes with plating. Gold plating on a component is very important. While tin plating is alright, gold plating lengthens the product life quite significantly. Plating is measured in microns. Some vendors get their first lot with gold plating but then move on to processes like flashing where gold is flashed. They then ship the products with a mix of flashed gold and gold plating. These may even pass the quality-control process, which is a problem.

Q. Any other elements that engineers should consider while selecting connectors or connector vendors?

A. Lead times for component delivery is important. As most busi-

nesses in this sector will tell you, a lead-time of a few days is good but a few weeks is not. Some vendors allow customers to buy without having to order based on a minimum order quantity. This ensures that the engineering team gets its components without having to face an unreasonable waiting period.

Q. What affects price changes?

A. Presence of gold in components tends to bring this interesting aspect of price change and revisions based on the price of gold. But some vendors are able to absorb small changes in gold prices and do not revise their components prices frequently. This helps the customers because they do not need to revise their bills of material cost every year to keep up with the price of gold. Interestingly, it is similar to what you see some major jewellers do.

Q. What are the latest technology trends in the world of connectors?

A. There are tie ups with competitors to buy and sell intellectual property to be able to serve customers better. Some of the strongest trends include miniaturisation from 1.27mm to 0.4mm pitch in ultra-micro interposers and interconnectors, as well as increased ruggedness of components. With an increase in demand for high-speed connectors, optics and connectors related to optics are another popular area. **EFY**